

ECE 212 Electronic Design Project 1 Report

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Introduction

The goal of this Electronic Design Project 1 was to build several foundational digital circuits that will be discussed and used throughout ECE 212. This project involves building a regulated power supply, creating LED indicators, learning to test circuits using the Analog Discovery 3 (AD3), and using several combinational logic circuits based on truth tables.

This report will describe the design, implementation, process, and testing of each circuit section. For each major portion of the project, the objective of the circuit, its design, and the behavior that is observed will be recorded and discussed. Project sections are shown below

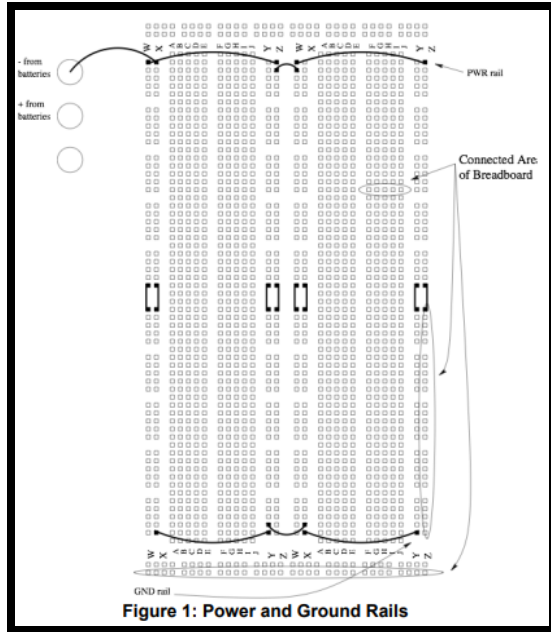
- **Power Test:** Sections 2
 - **LED Test:** Section 3
 - **LED Test with AD3:** Sections 4-5
 - **Logic Circuits Test:** Sections 7-11
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1. BreadBoard Setup

1.1 Objective

- Prepare the breadboard to construct a 5V voltage regulator that will power all digital logic circuits throughout the project.

1.2 Setup



The breadboard utilizes dedicated power rails along its edges for voltage distribution. The red rails serve as the positive supply (+5V), and the blue rails are for ground (GND). This arrangement ensures that all components on the board receive consistent power and ground connections.

Jumper wires were added as needed, as shown in **Figure 1**. Connecting all positive and negative rails along the length of both boards ensures that the power rail can be connected to +5V and ground from anywhere on the board.

2. Power Supply Circuit

2.1 Objective

- Build a regulated 5V power supply using the 7805 voltage regulator.

2.2 Circuit Design

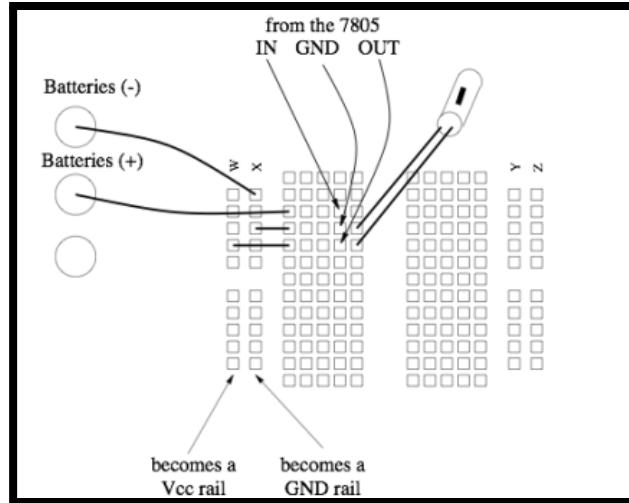
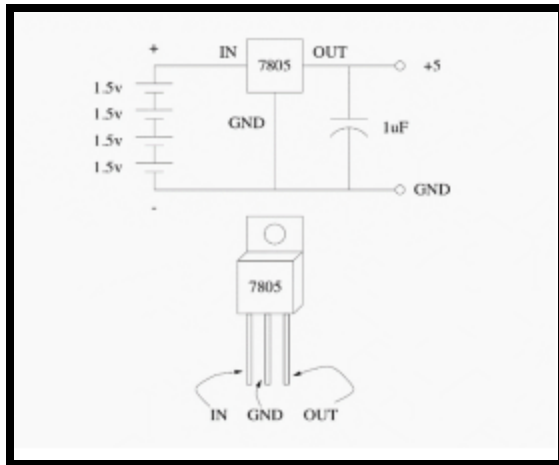


Figure 2: 7805 Voltage Regulator Diagram Figure 3: Layout of 7805 on Breadboard

The circuit uses a battery pack made up of four 1.5-volt batteries, which produces approximately 6V of input voltage. The battery pack's positive lead was fed into a 7805 voltage regulator, which stabilized the output voltage at 5V, making it ideal for TTL logic circuits.

The voltage regulator has three pins as described by the schematic in **Figure 2-3**:

- **Input** (connected to 6V battery pack)
- **Ground** (connected to GND rail)
- **Output** (regulated 5V)

The regulated output voltage is connected to the breadboard's power rail (+5V), and the ground pin is connected to the breadboard ground rail. These rails provide power to all components and serve as a ground.

2.3 Results/Verification

The power supply input was tested by connecting a 6V battery pack to a breadboard. Using a multimeter, the input voltage from the batteries was determined to be around 6 volts DC. The output voltage of the 7805 voltage regulator was then measured using the same method, resulting in approximately 5 volts DC. This confirmed that the regulator was working properly and that the circuit provided a stable voltage supply for TTL logic throughout the project.

3. LED Probe Circuit

3.1 Objective

- The purpose of the LED probe circuit was to create a simple diagnostic tool that could quickly determine whether a point in the circuit was at a logical HIGH or LOW state.

3.2 Circuit Design

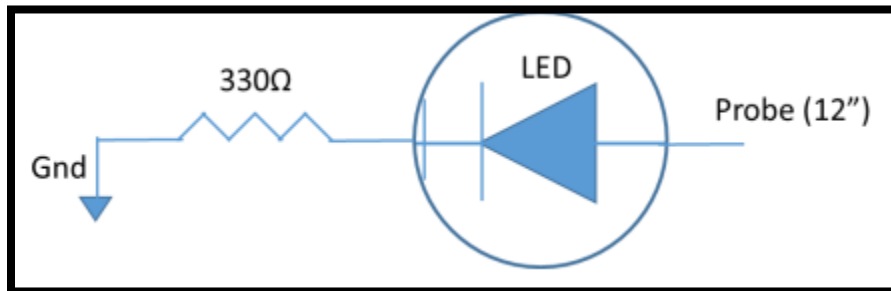


Diagram of the LED Probe Circuit

The LED Probe Circuit consisted of an LED connected in series with a 330-ohm resistor to limit the current flowing through the LED to prevent damage to the component. A long probe wire was connected to the circuit so that different points on the board could be tested.

3.3 Results

The LED probe was tested by connecting its wire to both the power rail (Vcc) and the ground rail (GND).

When connected to +5V, the LED lit up, indicating a logical HIGH signal. When connected to ground, the LED remained off, indicating a logical LOW. This probe circuit will function as a debugging tool for the duration of the project.

4. Buffered LED Indicator Circuits

4.1 Objective

- Construct an LED indicator circuit that could display digital output values produced by logic circuits

4.2 Circuit Design

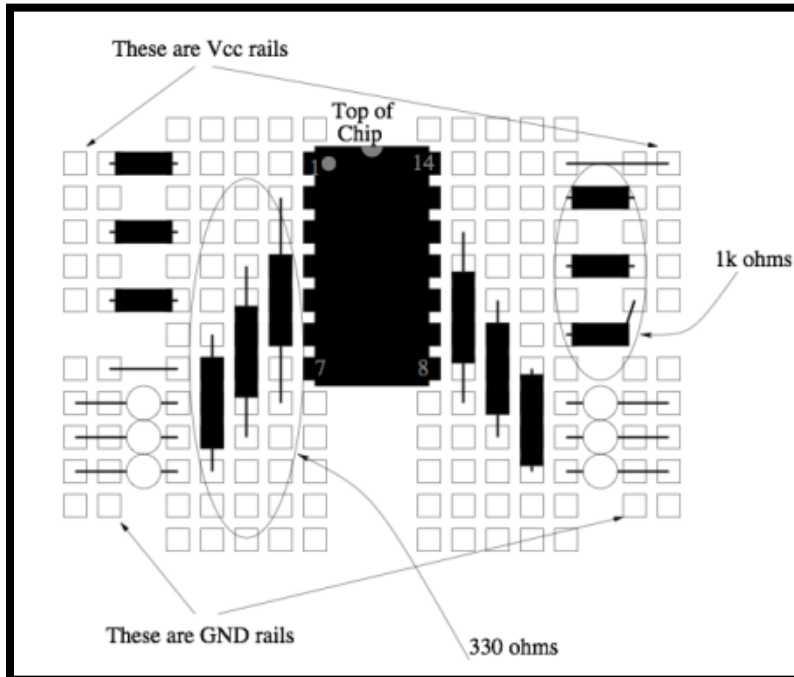


Figure 5: LED Setup on the Breadboard

The LED indicators were built using a 7405 hex inverter integrated circuit. The 7405 contains six independent inverter gates. Each LED indicator circuit consisted of:

- One inverter from the 7405 chip
- A 330-ohm current-limiting resistor
- A 1k-ohm pull-up resistor
- An LED

4.3 Testing Procedure

Each LED indicator was tested by manually connecting the inverter input pins to either the power rail (+5V) or the ground and debugged using the LED probe circuit.

4.4 Results

When the input of the inverter was connected to +5V, the inverter output produced a LOW signal, and the LED was off. When the input was connected to ground, the inverter output produced a HIGH signal, and the LED started to glow.

All LED indicator circuits acted as expected.

5. Testing LEDs with AD3 Static I/O

5.1 Objective

- Learn how to use the Analog Discovery 3 (AD3) and WaveForms software to control digital inputs and observe circuit outputs.

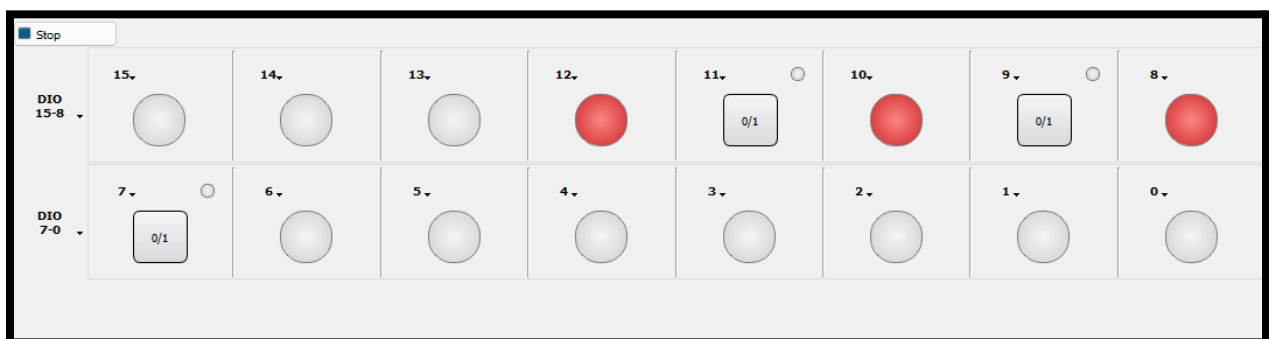
5.2 Setup

The Static I/O tool in WaveForms was used to manually control digital signals. Three digital outputs from the AD3 were configured as push buttons (DIO-7, 9, 11). These signals were connected to the inputs of three inverter circuits. Three additional digital channels were configured as LEDs to monitor the outputs of the inverter circuits (DIO-8, 10, 12).

The following connections were used:

AD3 Pin	Inverter Connection
DIO-7	Inverter Input to Pin 1
DIO-8	Inverter Output to Pin 2
DIO-9	Inverter Input to Pin 3
DIO-10	Inverter Output to Pin 4
DIO-11	Inverter Input to Pin 5
DIO-12	Inverter Output to Pin 6

5.3 Results



Screenshot of the Static I/O:

When a push button in the Static I/O interface was pressed, the corresponding inverter input changed state. The LED on the breadboard would respond immediately.

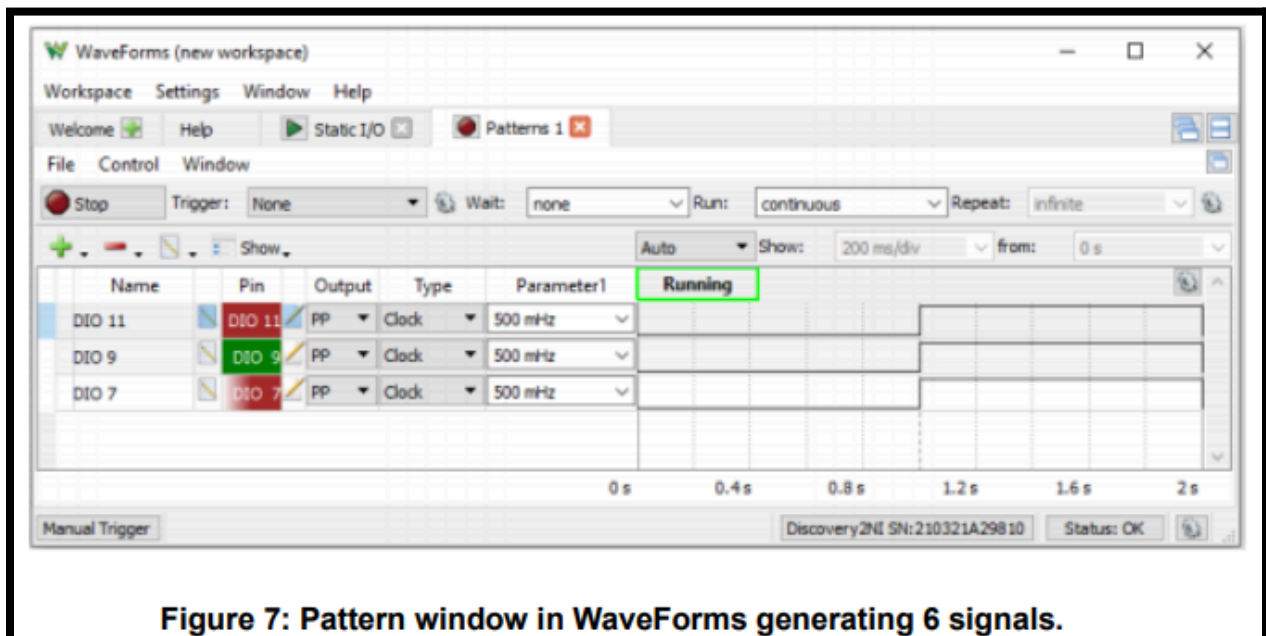
The LED displayed in the Static I/O interface also changed state, confirming that the output signal was correctly captured. Due to the inverter, the output signal appeared to be inverted relative to the input signal. This means the physical LED will have the opposite output as the LED display on WaveForms.

6. Pattern Generator and Logic Analyzer

6.1 Objective

- Generate automated input signals using the Pattern Generator in WaveForms and observe circuit behavior using the Logic Analyzer.

6.2 Pattern Generator Configuration



The Pattern Generator tool in WaveForms was used to create digital input signals that drive the circuit inputs. Three channels were configured to generate signals: DIO-7, DIO-9, and DIO-11.

Each signal was set up as a clock waveform at 500 mHz. This frequency was chosen to observe the signals alternate between logic HIGH and logic LOW about once every second. When the Pattern Generator ran, the AD3 produced these signals and drove the circuit inputs.

As the signals changed over time, the three LEDs connected to the inputs and outputs began to blink in response to a change in logic states. This process verified that the pattern generator was functioning correctly and that the proper inverter logic states were displayed on the board. The Pattern Generator was left running.

6.3 Logic Analyzer Setup

To observe the circuit's behavior while the Pattern Generator was running, the WaveForms logic analyzer tool was opened. The logic analyzer can monitor digital signals without affecting the actual circuit.

Six channels were added to the logic analyzer corresponding to the input and output signals.

- **DIO-7**—Input Signal (LED 1)
- **DIO-8**—Output Signal (LED 1)
- **DIO-9**—Input Signal (LED 2)
- **DIO-10**—Output Signal (LED 2)
- **DIO-11**—Input Signal (LED 3)
- **DIO-12**—Output Signal (LED 3)

The time base was set to 1 second per division (1 s/div) so that the relatively slow clock signals could be clearly observed. Once the logic analyzer was started, the signals were displayed as digital waveforms showing transitions between logical HIGH (1) and logical LOW (0).

6.4 Results

When the pattern generator and logic analyzer were running simultaneously, the logic analyzer showed repeated clock patterns in the input signals. The circuit outputs changed state in response to the waveform display.

The captured waveforms demonstrated that the pattern generator correctly generated digital clock signals and that the circuit outputs behaved as expected. The LEDs on the circuit blinked in sync with the waveform displayed in the logic analyzer, confirming that the hardware was correct.

7. Combinational Logic Circuits

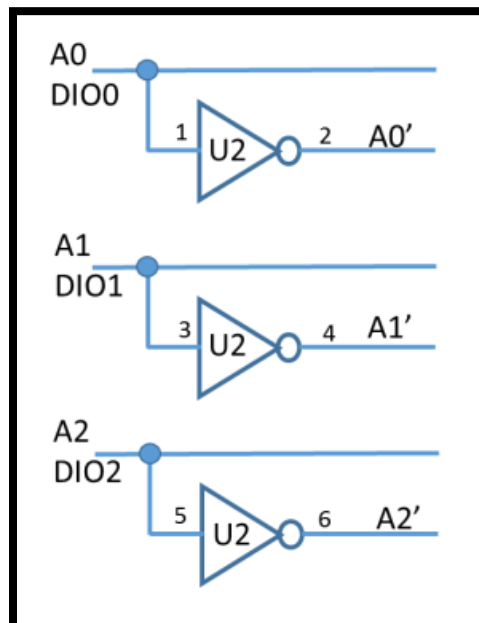
7.1 Objective

- Use the truth table provided below in **Table 3** to implement three combinational logic circuits—F2, F1, and F0—as well as their corresponding inputs, A2, A1, and A0.

A2	A1	A0	F2	F1	F0
0	0	0	0	1	1
0	0	1	0	0	1
0	1	0	0	0	0
0	1	1	1	0	0
1	0	0	0	1	0
1	0	1	1	1	0
1	1	0	1	1	1
1	1	1	0	1	1

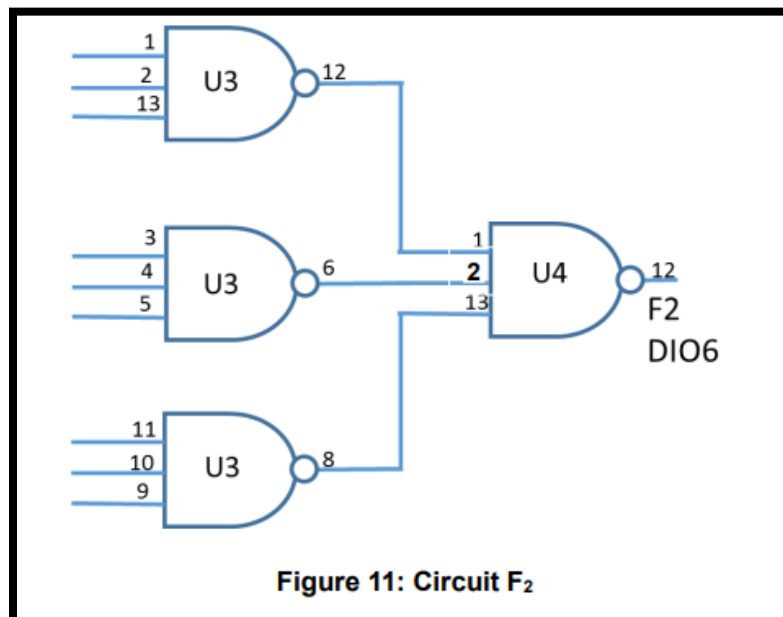
Table 3: Truth Table

A 7404 inverter was placed on the board as shown below, near the connection of the AD3 DIO pins, to create the A2', A1', and A0' outputs used to complete the truth table. These outputs will be used to complete the F2, F1, and F0 circuits.

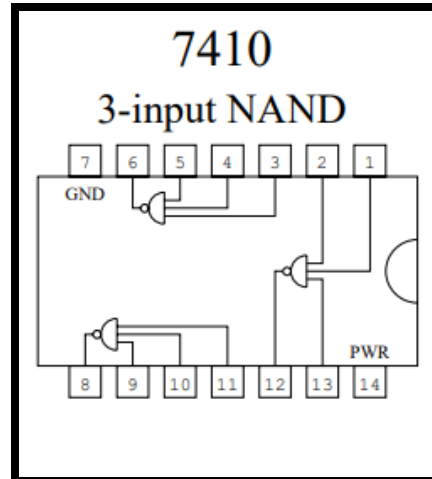


8. Circuit F2 Implementation

8.1 Design



The F2 circuit was built to produce the required F2 output for each combination of three inputs: A2, A1, and A0. The design was built using two 7410 3-input NAND ICs and tested by connecting them to the appropriate input and output lines. The final NAND gate output was connected to a fourth LED and read by DIO-6. The circuit diagram that was used is shown below.

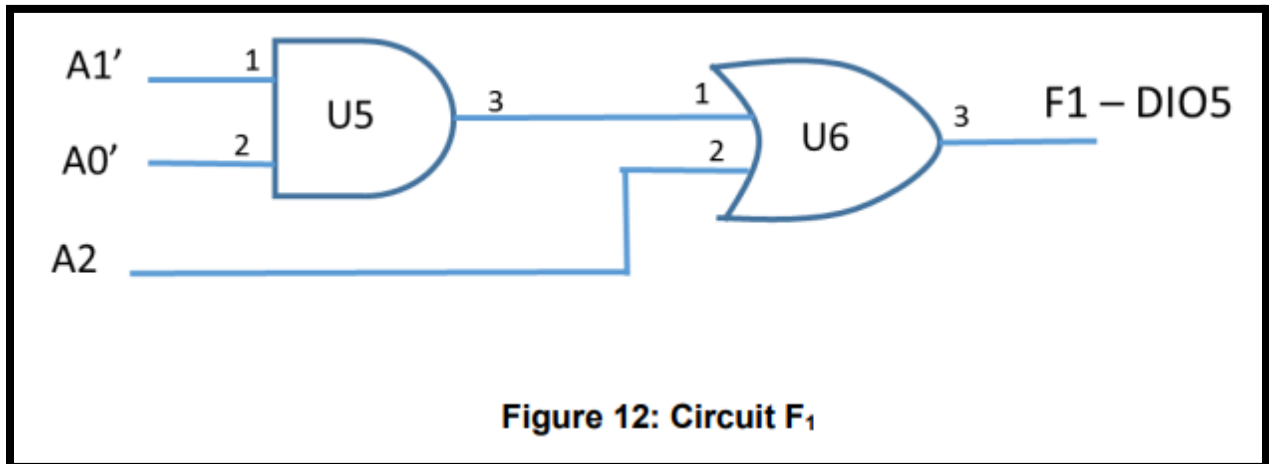


8.2 Results

After completing the circuit, the output of F2 was tested for all eight possible inputs using the Static I/O. The observed output was compared to the values specified in the truth table. The circuit returned the correct value for each input combination, indicating that the build was successful.

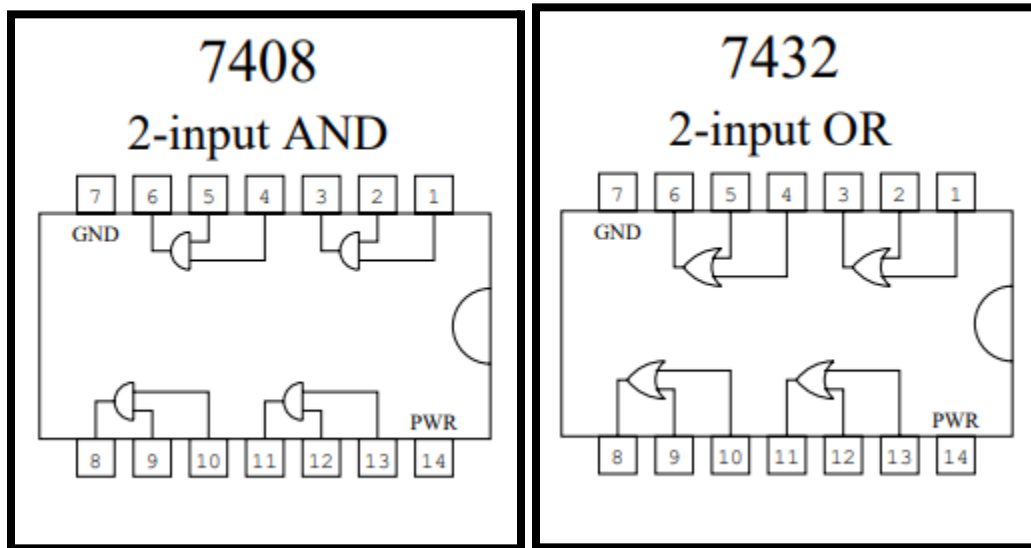
9. Circuit F1 Implementation

9.1 Design



The F1 circuit was built to produce the required F1 output for each combination of three inputs: A2, A1, and A0. The design was built using one 7408 2-input AND IC and one 7432 2-input OR

IC and tested by connecting them to the appropriate input and output lines. The output of F1 was connected to a fifth LED and read by DIO-5.

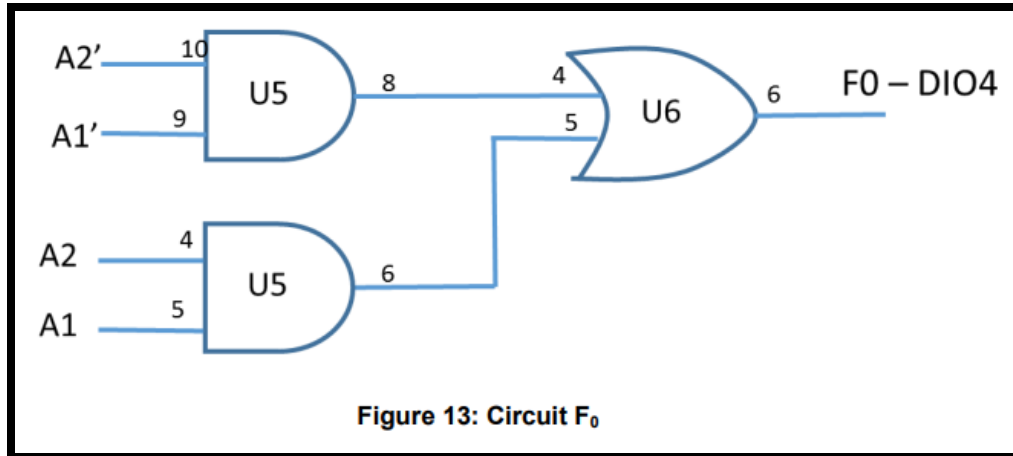


9.2 Results

After completing the circuit, the output of F1 was tested for all eight possible inputs using the Static I/O. The observed output was compared to the values specified in the truth table. The circuit returned the correct value for each input combination, indicating that the build was successful.

10. Circuit F0 Implementation

10.1 Design



The F0 circuit was built to produce the required F0 output for each combination of three inputs: A2, A1, and A0. The design was built using the same ICs as in the F1 circuit and tested by connecting them to the appropriate input and output lines. The output of F0 was connected to a sixth LED and read by DIO-4.

10.2 Results

After completing the circuit, the output of F0 was tested for all eight possible inputs using the Static I/O. The observed output was compared to the values specified in the truth table. The circuit returned the correct value for each input combination, indicating that the build was successful.

The Static I/O was used to verify that all combinational circuits worked simultaneously.

11. Binary Counter Test Using AD2 / AD3

11.1 Objective

- Use the WaveForms pattern generator to generate a binary input sequence, and then use the logic analyzer to verify the behavior of the combinational logic circuits.

11.2 Configuration

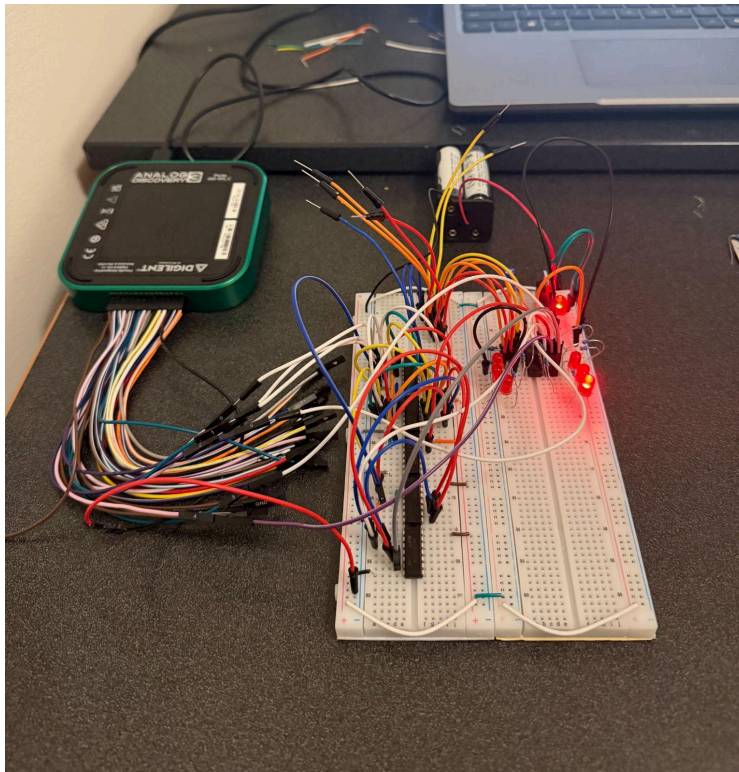
The AD3 was used to generate the uncomplemented input signals for the logic circuits. Inputs A2, A1, A0 were connected to DIO-0, DIO-1, and DIO-2, respectively, while the outputs F0, F1, and F2 were monitored on DIO-4, DIO-5, and DIO-6.

A bus was created in the WaveForms pattern generator using DIO-0, DIO-1, and DIO-2, with DIO-0 as the least significant bit and DIO-2 as the most significant bit. The bus was configured as a binary counter, which automatically generates the sequence 000 through 111.

11.3 Results

When the binary counter was running, the LEDs on the breadboard displayed the inverted outputs of F2, F1, and F0, while the logic analyzer displayed the true result to ensure that the outputs matched the expected values.

12. Completed Breadboard



The breadboard was designed to keep the circuit organized and simple to test, with jumper wires for easy placement and potential changes during the design process. The power supply and voltage regulator were placed near the top right corner of the board for easy access, while the 7405 inverter IC was placed directly beneath it on the opposite side of the AD3 probes and combinational circuits. Combinational circuits, AD3 probes, and the 7404 inverter were placed on the left side of the board to keep the LED configuration organized and reduce clutter.

13. Conclusion

In this project, the power supply, LED indicator circuits, and three combinational logic circuits were successfully built and tested on a breadboard. The circuits were tested on the AD3 with WaveForms software tools such as Static I/O, Pattern Generator, and Logic Analyzer. Testing the hardware with WaveForms confirmed that the outputs F2, F1, and F0 corresponded to the expected values from the truth table across all input combinations. Overall, the project showed correct circuit design and provided me with experience using digital logic and AD3 tools to test and verify the behavior of the circuit.